

Операционные системы

Продвинутое использование git

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Цели и задачи работы

Процесс выполнения лабораторной работы

Выводы по проделанной работе

1. Цели и задачи работы

Получение навыков правильной работы с репозиториями git.

- 1 Выполнить работу для тестового репозитория.
- 2 Преобразовать рабочий репозиторий в репозиторий с git-flow и conventional commits.

2. Процесс выполнения лабораторной работы

```
gakuligin@gakuligin:~$  
gakuligin@gakuligin:~$ pnpm setup  
Appended new lines to /home/gakuligin/.bashrc  
  
Next configuration changes were made:  
export PNPM_HOME="/home/gakuligin/.local/share/pnpm"  
case ":$PATH:" in  
  *:$PNPM_HOME:*) ;;  
  *) export PATH="$PNPM_HOME:$PATH" ;;  
esac  
  
To start using pnpm, run:  
source /home/gakuligin/.bashrc  
gakuligin@gakuligin:~$ source ~/.bashrc  
gakuligin@gakuligin:~$
```

Рисунок 1: Node.js

```
gakuligin@gakuligin:~$ pnpm add -g commitizen

Update available! 10.27.0 → 10.30.2.
Changelog: https://pnpm.io/v/10.30.2
To update, run: pnpm self-update

WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +149
+++++
Progress: resolved 149, reused 0, downloaded 149, added 149, done

/home/gakuligin/.local/share/pnpm/global/5:
+ commitizen 4.3.1

Done in 2.4s using pnpm v10.27.0
gakuligin@gakuligin:~$
```

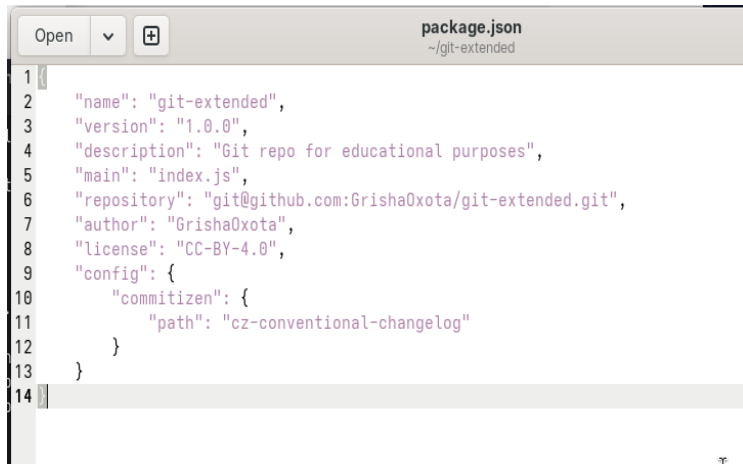
Рисунок 2: установка commitizen


```
gakuligin@gakuligin:~$ pnpm add -g standard-changelog
WARN 2 deprecated subdependencies found: glob@7.2.3, inflight@1.0.6
Packages: +32
+++++
Progress: resolved 181, reused 149, downloaded 32, added 32, done

/home/gakuligin/.local/share/pnpm/global/5:
+ standard-changelog 7.0.1

Done in 3s using pnpm v10.27.0
gakuligin@gakuligin:~$
```

Рисунок 3: установка standard-changelog



```
1 {  
2   "name": "git-extended",  
3   "version": "1.0.0",  
4   "description": "Git repo for educational purposes",  
5   "main": "index.js",  
6   "repository": "git@github.com:GrishaOxota/git-extended.git",  
7   "author": "GrishaOxota",  
8   "license": "CC-BY-4.0",  
9   "config": {  
10     "commitizen": {  
11       "path": "cz-conventional-changelog"  
12     }  
13   }  
14 }
```

Рисунок 4: package.json

```
gakuligin@gakuligin:~/git-extended$ git add .
gakuligin@gakuligin:~/git-extended$ git cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't
modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master ffb06fe] chore: test
 1 file changed, 14 insertions(+)
 create mode 100644 package.json
gakuligin@gakuligin:~/git-extended$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.12 KiB | 1.12 MiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:GrishaOxota/git-extended.git
 29ba196..ffb06fe  master -> master
gakuligin@gakuligin:~/git-extended$
```

Рисунок 5: Отправка

```
gaku@gaku:~/git-extended$ git flow init
Which branch should be used for bringing forth production releases?
- master
Branch name for production releases: [master]
Branch name for "next release" development: [develop]

How to name your supporting branch prefixes?
Feature branches? [feature/]
Bugfix branches? [bugfix/]
Release branches? [release/]
Hotfix branches? [hotfix/]
Support branches? [support/]
Version tag prefix? [] v
Hooks and filters directory? [/home/gakuligin/git-extended/.git/hooks]
gaku@gaku:~/git-extended$ git branch
* develop
  master
gaku@gaku:~/git-extended$ git push --all
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'develop' on GitHub by visiting:
remote:   https://github.com/Grisha0xota/git-extended/pull/new/develop
remote:
To github.com:Grisha0xota/git-extended.git
 * [new branch]      develop -> develop
gaku@gaku:~/git-extended$
```

Рисунок 6: Инициализация и загрузка изменений

```
gaku@gaku:~/git-extended$ git branch --set-upstream-to=origin/develop develop
branch 'develop' set up to track 'origin/develop'.
gaku@gaku:~/git-extended$ git flow release start 1.0.0
Switched to a new branch 'release/1.0.0'

Summary of actions:
- A new branch 'release/1.0.0' was created, based on 'develop'
- You are now on branch 'release/1.0.0'

Follow-up actions:
- Bump the version number now!
- Start committing last-minute fixes in preparing your release
- When done, run:

    git flow release finish '1.0.0'

gaku@gaku:~/git-extended$ standard-changelog --first-release
gaku@gaku:~/git-extended$ git add CHANGELOG.md
gaku@gaku:~/git-extended$ git commit -am 'chore(site): add changelog'
[release/1.0.0 5c2e274] chore(site): add changelog
1 file changed, 4 insertions(+)
create mode 100644 CHANGELOG.md
gaku@gaku:~/git-extended$ git flow release finish 1.0.0
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
Merge made by the 'ort' strategy.
CHANGELOG.md | 4 +++
1 file changed, 4 insertions(+)
create mode 100644 CHANGELOG.md
```

Рисунок 7: Завершение релиза

Отправим данные на github

```
gkuligin@gkuligin:~/git-extended$ git push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 2.67 KiB | 2.67 MiB/s, done.
Total 5 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:GrishaOxota/git-extended.git
   ffb06fe..97c06c5  develop -> develop
   ffb06fe..00623c9  master -> master
gkuligin@gkuligin:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 159 bytes | 159.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:GrishaOxota/git-extended.git
 * [new tag]          v1.0.0 -> v1.0.0
gkuligin@gkuligin:~/git-extended$ gh release create v1.0.0 -F CHANGELOG.md
https://github.com/GrishaOxota/git-extended/releases/tag/v1.0.0
gkuligin@gkuligin:~/git-extended$
```

Рисунок 8: Отправка

```
gakuligin@gakuligin:~/git-extended$ git flow feature start feature_branch
Switched to a new branch 'feature/feature_branch'

Summary of actions:
- A new branch 'feature/feature_branch' was created, based on 'develop'
- You are now on branch 'feature/feature_branch'

Now, start committing on your feature. When done, use:

    git flow feature finish feature_branch

gakuligin@gakuligin:~/git-extended$ touch test2
gakuligin@gakuligin:~/git-extended$ git flow feature finish feature_branch
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.
Already up to date.
Deleted branch feature/feature_branch (was 97c06c5).

Summary of actions:
- The feature branch 'feature/feature_branch' was merged into 'develop'
- Feature branch 'feature/feature_branch' has been locally deleted
- You are now on branch 'develop'

gakuligin@gakuligin:~/git-extended$
```

Рисунок 9: Объединение веток

```
Summary of actions:
- Release branch 'release/1.2.3' has been merged into 'master'
- The release was tagged 'v1.2.3'
- Release tag 'v1.2.3' has been back-merged into 'develop'
- Release branch 'release/1.2.3' has been locally deleted
- You are now on branch 'develop'

gakuligin@gakuligin:~/git-extended$ git push --all
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 8 threads
Compressing objects: 100% (6/6), done.
Writing objects: 100% (6/6), 2.78 KiB | 2.78 MiB/s, done.
Total 6 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:GrishaOxota/git-extended.git
   97c06c5..0a55551  develop -> develop
   00623c9..58990ec  master -> master
gakuligin@gakuligin:~/git-extended$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 160 bytes | 160.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:GrishaOxota/git-extended.git
 * [new tag]         v1.2.3 -> v1.2.3
gakuligin@gakuligin:~/git-extended$ gh release create v1.2.3 -F CHANGELOG.md
https://github.com/GrishaOxota/git-extended/releases/tag/v1.2.3
gakuligin@gakuligin:~/git-extended$
```

Рисунок 10: Завершение релиза


```
gakuligin@gakuligin:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git
cz
cz-cli@4.3.1, cz-conventional-changelog@3.3.0

? Select the type of change that you're committing: chore:    Other changes that don't
modify src or test files
? What is the scope of this change (e.g. component or file name): (press enter to skip)
? Write a short, imperative tense description of the change (max 93 chars):
  (4) test
? Provide a longer description of the change: (press enter to skip)

? Are there any breaking changes? No
? Does this change affect any open issues? No
[master 3550f79] chore: test
 1 file changed, 11 insertions(+), 11 deletions(-)
gakuligin@gakuligin:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git
push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 1.11 KiB | 1.11 MiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Grisha0xota/2026-1--study--os-intro.git
 d44da2f..3550f79  master -> master
gakuligin@gakuligin:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 11: package.json и коммит

```
Summary of actions:
- Release branch 'release/1.0.0' has been merged into 'master'
- The release was tagged 'v1.0.0'
- Release tag 'v1.0.0' has been back-merged into 'develop'
- Release branch 'release/1.0.0' has been locally deleted
- You are now on branch 'develop'

gakuligin@gakuligin:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git
push --all
Enumerating objects: 6, done.
Counting objects: 100% (6/6), done.
Delta compression using up to 8 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 2.76 KiB | 2.76 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:Grisha0xota/2026-1--study--os-intro.git
   3550f79..841940f  develop -> develop
   3550f79..182aa80  master -> master
gakuligin@gakuligin:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$ git
push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 158 bytes | 158.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To github.com:Grisha0xota/2026-1--study--os-intro.git
 * [new tag]          v1.0.0 -> v1.0.0
gakuligin@gakuligin:~/work/study/2025-2026/Операционные системы/2026-1--study--os-intro$
```

Рисунок 12: Завершение релиза

3. Выводы по проделанной работе

Мы приобрели практические навыки взаимодействия с дополнительными функциями гитхаб.